Homework Week 7

Modern Physics

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(Dated: April 7, 2026)

For exercises from the textbook, the corresponding number is reported. The text may have been edited for clarity.

Some of the following exercises (possibly with different numbers) may be on the **mini-test** taking place on the Monday of the following week, or on the **midterms**.

THE HYDROGEN ATOM, ANGULAR MOMENTUM, NORMAL ZEEMAN EFFECT

- **L_z for Hydrogen excited state / Ex. 7.19**
  Calculate the possible z components of the orbital angular momentum for an electron in a 3p state.

- **Hydrogen atom in a magnetic field / Ex. 7.25**
  A hydrogen atom in an excited 5f state is in a magnetic field $B = 3.00\,\mathrm{T}$. (a) How many energy states can the electron have in the 5f subshell? (Ignore the magnetic spin effects). (b) What is the energy of the 5f state in the absence of a magnetic field? (c) What will be the energy of each state in the magnetic field?

- **Angular momentum conservation of the Hydrogen atom**
  Check that L_z and $\mathbf{L}^2$ commute with the Hamiltonian of the Hydrogen atom using polar coordinates:

$$L_z = i\hbar\frac{\partial}{\partial\phi},$$

$$L^2 = -\hbar^2\left(\frac{1}{\sin^2\theta}\frac{\partial^2}{\partial\phi^2} + \frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right)\right).$$

  Do the L_x and L_y components commute with H?

- **Solve the rest of problems 7.19 - 7.27 from the book**

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EXTRA QUANTUM MECHANICS PROBLEMS

These are more advanced problems that are above the level we expect on the midterms, but are very useful to solve as practice. If you have any questions, feel free to contact the instructors.

- **Angular momentum representations with finite matrices**
  So far, we have only considered angular momentum operators as differential operators. However, it is possible to find sets of finite matrices that also satisfy the same commutation relations, in technical terms, they form an *angular momentum representation*. This is a consequence of the fact that for a fixed l there is only a finite number of L_z eigenstates with $m = -l, \ldots, l$, giving $2l+1$ possible values in total. Be aware that such representation is not unique.

  For each of the following sets of matrices: (i) Check that L_x, L_y, and L_z satisfy the angular momentum commutation relations. (ii) Find the eigenvalues m and eigenvectors $|m\rangle$ of L_z. (iii) Compute $\mathbf{L}^2$ and show that it commutes with all L components. (iv) Find the value of l from the relation that the eigenvalue of $\mathbf{L}^2$ equals $\hbar^2 l(l+1)$, and verify that the possible values of m are $m = -l, \ldots, l$.

  (a)
$$L_x = \frac{\hbar}{\sqrt{2}} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \qquad L_y = \frac{\hbar}{\sqrt{2}} \begin{pmatrix} 0 & -i & 0 \\ i & 0 & -i \\ 0 & i & 0 \end{pmatrix} \qquad L_z = \hbar \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}$$

  (b)
$$L_x = i\hbar \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{pmatrix} \qquad L_y = i\hbar \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{pmatrix} \qquad L_z = i\hbar \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

  (c)
$$L_x = \frac{\hbar}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \qquad L_y = \frac{\hbar}{2} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \qquad L_z = \frac{\hbar}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

- **Angular momentum ladder operators**
  It is possible to define so called "ladder" operators for the angular momentum z component, these take an eigenstate of L_z to another eigenstate with one $\hbar$ unit higher or lower eigenvalue. The ladder operators are defined as

$$L_\pm = L_x \pm i L_y.$$

  Prove the commutation relation

$$[L_z, L_\pm] = \hbar L_\pm.$$

  Show that, as a consequence of this commutation relation, if $|m\rangle$ is an eigenstate of L_z with eigenvalue $\hbar m$, then $L_\pm|m\rangle$ is a (not necessarily normalized) eigenstate of L_z with eigenvalue $m' = m \pm \hbar$. (i.e., if a state $|m\rangle$ satisfies $L_z|m\rangle = \hbar m|m\rangle$, then the state $|m'\rangle = L_\pm|m\rangle$ satisfies $L_z|m'\rangle = \hbar m'|m'\rangle$ with $m' = m \pm 1$.)

  Write down the ladder operators explicitly for the angular momentum matrices in the previous problem, and check that they take L_z eigenstates into other eigenstates.

  Bonus: Using explicit calculation with the differential operator form of L, show that for the hydrogen 2p orbitals $L_\pm \psi_{210} = \sqrt{2}\psi_{21\pm1}$. What is $L_+\psi_{211}$?

  Bonus: We saw that $|m \pm 1\rangle = \alpha_\pm L_\pm |m\rangle$ with some normalization constant $\alpha_\pm$ such that both $|m\rangle$ and $|m \pm 1\rangle$ are normalized. Find the value of $\alpha_\pm$.

  Hint: First show that $(L_\pm)^\dagger = L_\mp$, $L_\mp L_\pm = \mathbf{L}^2 - (L_z^2 \pm L_z)$, and use that $\mathbf{L}^2|m\rangle = \hbar^2 l(l+1)|m\rangle$. You will find that $\alpha_\pm$ depends on both l and m.